Algebra & Functions — Timed Simulation 2

IBEW / ETA math section format

33	46	~84 sec	No
questions	minutes	per question	calculator

About this test

- All 33 questions from your photos of Timed Simulation 2, retyped clean and printable, in the original order. Wording, numbers and answer choices are kept as they appeared on screen.
- Graphs are redrawn on a clean grid to match what was on screen.
- Multiple choice, A–D. No calculator. No penalty for guessing — answer everything.

How to take it

- Set a timer for 46 minutes and don't stop it.
- Skip anything that takes more than about 90 seconds and come back to it.
- The answer key with a short explanation for every question is at the back.

What's on it

Simplifying, expanding & factoring	Questions 9, 10, 20, 22, 23, 31
Number series	Questions 2, 5, 11, 16, 19, 29
Plugging in & solving	Questions 6, 12, 13, 17, 21, 24, 32
Isolating a variable	Questions 8, 18, 26
Variable relationships (which statement is true)	Questions 1, 3, 4, 15, 33
Same relationship: tables, graphs, words	Questions 7, 14, 28, 30
Graphing & systems	Questions 25, 27

Question 1

Consider the following formula:

$$S = \frac{N}{2} + 8$$

Which of the following statements is true for this formula?

A) When the value of N is less than -16, S is positive.

B) When the value of N is greater than 16, S is negative.

C) When the value of N is greater than -16, S is negative.

D) When the value of N is less than -16, S is negative.

WORKSPACE

Question 2

What is the next number in the following series of numbers?

4 7 8 11 13 16 ?

A) 18

B) 21

C) 17

D) 19

WORKSPACE

Question 3

Consider the following formula:

$M + 2N = 0$

Which of the following statements is true for this formula?

A) M always has a greater value than N.

B) M always has a lower value than N.

C) M is always equal to N.

D) M may be either greater or lower than N.

WORKSPACE

Question 4

Consider the following formula:

$$M = N \div \frac{1}{2}$$

Which of the following statements is true for this formula (assume M and N are different than zero)?

A) $MN > N^2$

B) $N + 2 > M$

C) $2N > M$

D) $M + \frac{1}{2} > N$

WORKSPACE

Question 5

What is the next number in the following series of numbers?

$-1\ 2\ 1\ 3\ 4\ ?$

A) 5

B) 7

C) 1

D) 6

WORKSPACE

Question 6

Solve the system:

(I) $5x - 2y = -7$

(II) $y - 5x = 6$

A) (1, 5)

B) $(-1, -5)$

C) $(-4, -1)$

D) $(-1, 1)$

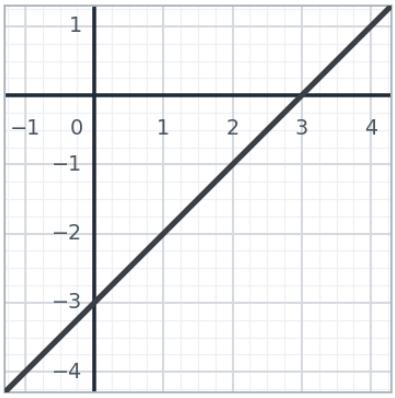
WORKSPACE

Question 7

Consider the following information:

Y is equivalent to the difference between twice the value of X and a constant C, where C equals 3.

Which of the following options represents the same relationship as the aforementioned information?

A) 

B) X is equivalent to the sum of half the value of Y and a constant C, where C equals 3.

C) $Y = 2X + 3$

D)

x	y
0	−3
1	−1
2	1
3	3
4	5

WORKSPACE

Question 8

Isolate the variable from the formula:

$E = mc^2$

$c = ?$

A) $\sqrt{\frac{E}{m}}$

B) $\sqrt{\frac{mc}{E}}$

C) $\sqrt{\frac{E}{mc}}$

D) $\sqrt{\frac{m}{E}}$

WORKSPACE

Question 9

Simplify the expression:

$(x^3 + 2)(x^3 - x^2) - x^3 + x^4$

A) $x^9 + x^6 + x^4 + 2x^3 - x^2$

B) $x^6 - x^5 + x^4 + x^3 - 2x^2$

C) $x^6 - x^5 + 2x^3 - 2x^2$

D) $x^6 + x^5 - x^3 - 2x^2$

WORKSPACE

Question 10

Simplify the following expression to its simplest form.

Consider $A = 7$ and $C = 5$:

$$B(C + 3) + A(2B - (B - 5))$$

A) $B - 35$

B) $-2B^2 - 62B$

C) $15B + 35$

D) $10B - 2$

WORKSPACE

Question 11

What is the next number in the following series of numbers?

50 26 14 8 5 ?

A) −1.5

B) 3.5

C) 2

D) 1.5

WORKSPACE

Question 12

Consider the following formula:

$$5(A+B)-3=C(B+C)$$

If $A=5$ and $C=2$, what is the value of B?

A) 3

B) -3

C) 6

D) -6

WORKSPACE

Question 13

Evaluate the value of the unknown variable:

$$\frac{a}{bc} - \frac{d(a-c)}{e+g} = J$$

$a = 24,\ b = -2,\ c = 4,\ d = -3,\ e = 7$, and $g = 3$. Find J.

A) 9

B) -9

C) -3

D) 3

WORKSPACE

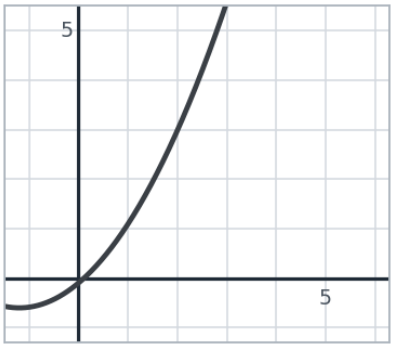
Question 14

Consider the following table:

x	y
0	−1
1	1
2	3
3	5
4	7
5	9

Which of the following choices represents the same relationship as demonstrated in this table?

A) Y is equivalent to twice the value of X and a constant C, where C equals −1.

B) 

C) $x = \frac{y - 1}{2}$

D)

x	y
−1	−1
−2	−3
−3	−5
−4	−7
−5	−9

WORKSPACE

Question 15

Consider the following formula:

$$2A = 6 - \frac{B}{3}$$

Which of the following statements is true for this formula?

A) If A is less than 3, B is positive.

B) If A is less than 1, B is negative.

C) If A is greater than 1, B is positive.

D) If A is greater than 1, B is negative.

WORKSPACE

Question 16

What is the next number in the following series of numbers?

39 27 18 12 9 ?

A) 6

B) 9

C) 12

D) 7.5

WORKSPACE

Question 17

Consider the following formula:

$AB = A + B + 5$

If $B = 4$, which of the following statements is true for this formula?

A) $A > B$

B) $A = B$

C) $A - B > 0$

D) $B - A > 0$

WORKSPACE

Question 18

Isolate the variable from the formula:

$y = (x - 4)^2 + 7$

$x = ?$

A) $y - 3$

B) $4 + \sqrt{y - 7}$

C) $-4\sqrt{y - 7}$

D) $\sqrt{y - 3}$

WORKSPACE

Question 19

What is the next number in the following series of numbers?

1 1 2 4 6 18 21 ?

A) 84

B) 63

C) 26

D) 24

WORKSPACE

Question 20

Factor the polynomial:

$x^5y^2z^6 - x^3y^2z^7$

A) $x^3yz^6(x^2y - z)$

B) $x^5y^2z^5(y - z^2)$

C) $x^5y^2z^7(x^{-2} - z)$

D) $x^3y^2z^6(x^2 - z)$

WORKSPACE

Question 21

Consider the following formula:

$ABC - 4 = A(B + C - 1)$

If $B = 2$ and $C = 3$, what is the value of A?

A) 2

B) 4

C) -2

D) -4

WORKSPACE

Question 22

Turn the following expression into a simple polynomial:

$4(-x^2 - 1) - x^2(2 - 6x)$

A) $-6x^3 + 2x^2 - 4$

B) $6x^3 - 5x^2 - 4x$

C) $-6x^3 - 5x^2 - 16$

D) $6x^3 - 6x^2 - 4$

WORKSPACE

Question 23

Which of the following functions are equivalent to one another (yield the same values)?

A. $x + 2$ B. $2x + 1$ C. $\frac{x + x + 2}{2}$ D. $-(-x - x - 1)$

A) A & C

B) A & D

C) B & C

D) B & D

WORKSPACE

Question 24

Evaluate the value of the unknown variable:

$v_i = 60 \qquad a = 6 \qquad t = 7.5$

$v_f = v_i + at$

$v_f = ?$

A) 95

B) 98

C) 102

D) 105

WORKSPACE

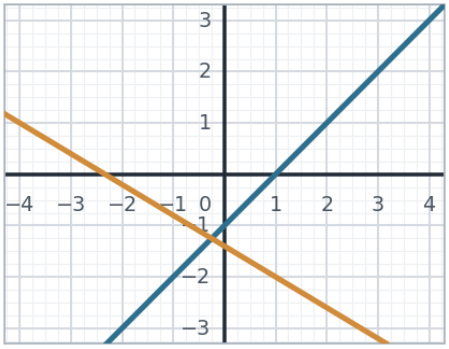
Question 25

Which of the graphs represent the following two linear equations?

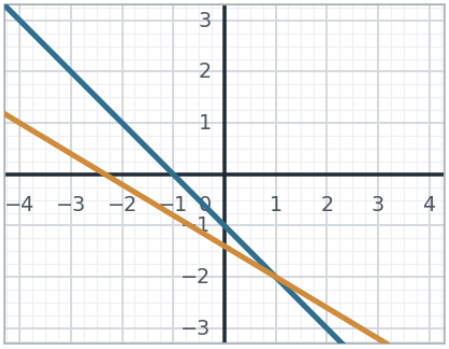
$4x + 4y = 4$

$-3x - 5y = -7$

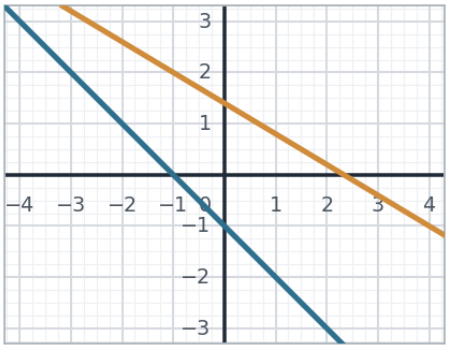
A)



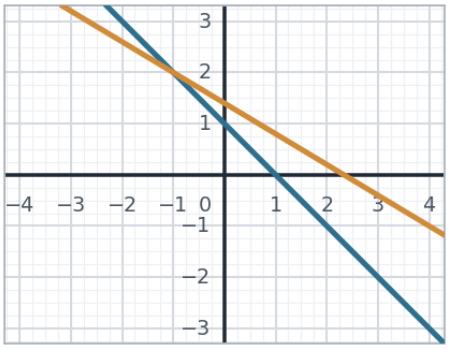
B)



C)



D)



WORKSPACE

Question 26

Isolate the variable from the formula:

$$x = 9 + \frac{1}{3(n-3)}$$

$n = ?$

A) $\frac{1}{3(x-9)} + 3$

B) $\frac{-2}{x-9} + 3$

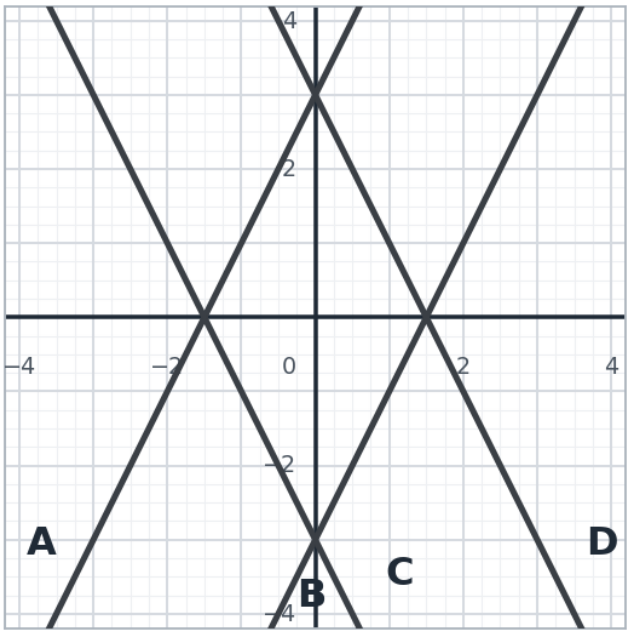
C) $\frac{1}{x-3} + 3$

D) $3x - 24$

WORKSPACE

Question 27

Which of the following graphs represents the function $y = -2x + 3$?



A) A

B) B

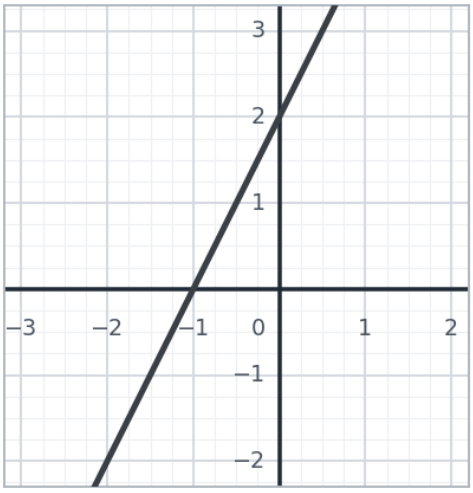
C) C

D) D

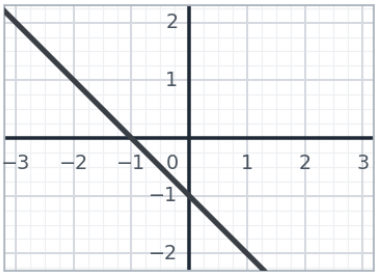
WORKSPACE

Question 28

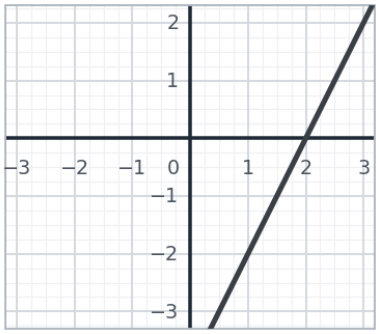
Consider the following graph of the function $y = mx + b$:



Graph B



Graph D



Which of the following options is true in regard to this graph?

A) Function A, $y = -\frac{1}{2}x + 2$, has the same slope and the same interception point with the y-axis.

B) Graph B is of a function with a different slope m and the same interception point with the y-axis.

C) Function C, $y = -2x - 1$, has a different slope and the same interception point with the y-axis.

D) Graph D is of a function with the same slope m and a different interception point with the y-axis.

Question 29

What is the next number in the following series of numbers?

1 4 9 16 25 ?

A) 37

B) 38

C) 36

D) 40

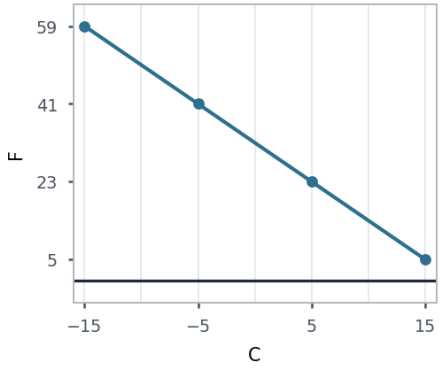
WORKSPACE

Question 30

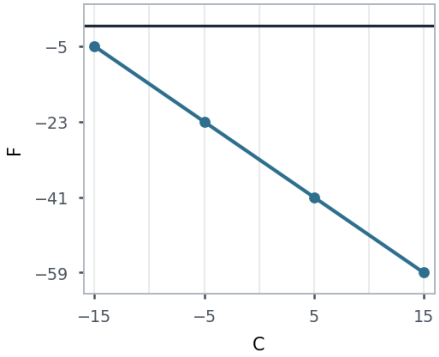
Which of the graphs represents the following table of values?

x	y
−15	5
−5	23
5	41
15	59

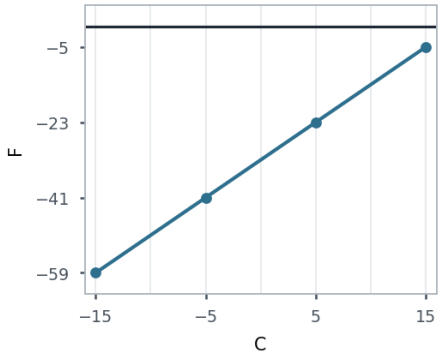
A)



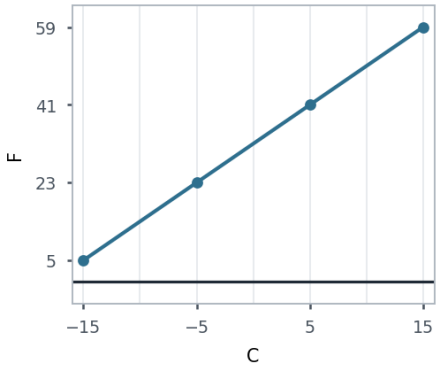
B)



C)



D)



WORKSPACE

Question 31

Factor the polynomial:

$x^2 - 4x + 4$

A) $(x - 1)(x - 4)$

B) $(x + 2)^2$

C) $(x - 2)^2$

D) $(x - 1)(x + 4)$

WORKSPACE

Question 32

Solve the proportion:

$$\frac{9}{5} = \frac{16y + 5}{9y - 6}$$

A) 79

B) 54

C) 29

D) 25

WORKSPACE

Question 33

Consider the following formula:

$3D = E - 3$

Which of the following statements is true for this formula?

A) If D is less than -1, E is positive.

B) If D is greater than -3, E is negative.

C) If D is greater than -1, E is positive.

D) If D is greater than -3, E is positive.

WORKSPACE

Answer Key

1. D	**10. C**	**19. A**	**28. D**
2. D	**11. B**	**20. D**	**29. C**
3. D	**12. D**	**21. A**	**30. D**
4. A	**13. D**	**22. D**	**31. C**
5. B	**14. A**	**23. D**	**32. A**
6. D	**15. A**	**24. D**	**33. C**
7. D	**16. B**	**25. D**	
8. A	**17. D**	**26. A**	
9. B	**18. B**	**27. D**	

1. D $S = 0$ when $\frac{N}{2} = -8$, so $N = -16$. Below -16, S is negative ✓

2. D Two patterns woven together: 4, 8, 13 (+4, +5) and 7, 11, 16 (+4, +5).
Next belongs to the first pattern, adding 6: $13 + 6 = 19$

3. D $M = -2N$. If $N = 1$, $M = -2$ (lower). If $N = -1$, $M = 2$ (greater).

4. A Dividing by $\frac{1}{2}$ doubles: $M = 2N$. $MN = 2N^2$, always bigger than N^2 ✓
C: $2N = M$, not greater. D fails at $N = -1$ ($-1.5 > -1$ is false).

5. B Each number is the sum of the two before it: $-1 + 2 = 1$, $2 + 1 = 3$, $1 + 3 = 4$, $3 + 4 = 7$

6. D From II: $y = 5x + 6$. Plug in: $5x - 10x - 12 = -7 \rightarrow -5x = 5 \rightarrow x = -1$, $y = 1$

7. D $Y = 2X - 3$. Every row of the table fits: $2(0) - 3 = -3$, $2(4) - 3 = 5$ ✓
The graph has slope 1 ($y = x - 3$). B works out to $Y = 2X - 6$.

8. A Divide by m: $c^2 = \frac{E}{m}$. Square root both sides.

9. B FOIL: $x^6 - x^5 + 2x^3 - 2x^2$. Then $-x^3 + x^4$: $x^6 - x^5 + x^4 + x^3 - 2x^2$

10. C $8B + 7(2B - B + 5) = 8B + 7(B + 5) = 8B + 7B + 35 = 15B + 35$

11. B Differences: -24, -12, -6, -3 (each half the last). Next is -1.5: $5 - 1.5 = 3.5$

12. D $25 + 5B - 3 = 2B + 4 \rightarrow 3B = -18 \rightarrow B = -6$

13. D $\frac{24}{-8} = -3$. $\frac{-3(20)}{10} = -6$. $J = -3 - (-6) = 3$

Answer Key (cont.)

14. A Starts at -1, goes up 2 each step: $y = 2x - 1$. A says exactly that ✓
C is $y = 2x + 1$. Table D is $y = 2x + 1$ too.

15. A Solve for B: $B = 18 - 6A$. $B = 0$ at $A = 3$, and below 3 B is positive ✓

16. B Differences: -12, -9, -6, -3 (each 3 smaller). Next is 0: $9 + 0 = 9$

17. D $4A = A + 9 \rightarrow 3A = 9 \rightarrow A = 3$. $B - A = 1 > 0$ ✓

18. B Subtract 7: $y - 7 = (x - 4)^2$. Square root: $\sqrt{y - 7} = x - 4$. Add 4.

19. A Alternates multiply then add with a growing number: $\times 1$, $+1$, $\times 2$, $+2$, $\times 3$, $+3$, $\times 4$
$21 \times 4 = 84$

20. D Pull out the smallest powers: x^3, y^2, z^6. What's left: $x^2 - z$

21. A $6A - 4 = 4A \rightarrow 2A = 4 \rightarrow A = 2$

22. D $-4x^2 - 4 - 2x^2 + 6x^3 = 6x^3 - 6x^2 - 4$

23. D C simplifies to $x + 1$. D simplifies to $2x + 1$, the same as B.

24. D $60 + 6 \times 7.5 = 60 + 45 = 105$

25. D Get y alone: $y = 1 - x$ and $y = \frac{7}{5} - \frac{3}{5}x$.
Blue crosses the y-axis at 1, orange at 1.4, both falling. They meet at $(-1, 2)$. That's D.

26. A Subtract 9: $x - 9 = \frac{1}{3(n-3)}$. Flip: $3(n - 3) = \frac{1}{x - 9}$.
Divide by 3, add 3: $n = \frac{1}{3(x-9)} + 3$

27. D Look for the line crossing the y-axis at 3 and falling 2 per 1.
A also crosses at 3 but rises. D is the falling one.

28. D Given line: crosses at $(-1, 0)$ and $(0, 2) \rightarrow y = 2x + 2$.
Graph D rises 2 per 1 (same slope) but crosses the y-axis lower ✓
A and C have a different slope and different y-intercept. Graph B crosses at -1, not 2.

29. C Perfect squares: $1^2, 2^2, 3^2, 4^2, 5^2$ … next is $6^2 = 36$

Answer Key (cont.)

30. D As C goes up, F goes up (from 5 to 59), and all F values are positive.
Only graph D rises through positive values. (This is the Celsius → Fahrenheit formula.)

31. C Two numbers that multiply to 4 and add to −4: −2 and −2. So $(x-2)^2$.

32. A Cross-multiply: $81y - 54 = 80y + 25 \rightarrow y = 79$

33. C $E = 3D + 3$, which is 0 at $D = -1$. Above −1, E is positive ✓
D fails at $D = -2$ ($E = -3$).